

PRSV: An Improved Peng–Robinson Equation of State for Pure Compounds and Mixtures

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The temperature and acentric factor dependence of the attractive term of the Peng–Robinson equation of state have been modified. The introduction of a single pure compound parameter allows the accurate reproduction of the vapor pressure data for a wide variety of substances. Nonpolar, polar nonassociating and associating compounds are equally well represented by the cubic PRSV equation of state. The conventional one-binary-parameter mixing rule allows the correlation of the vapor–liquid equilibrium data for a wide variety of binary systems. Only for systems formed by a polar compound (associating or not) and a saturated hydrocarbon, are results poorer than those obtained with conventional excess Gibbs energy functions.

On a modifié la dépendance en température et en facteur d'acentricité du terme attractif de l'équation d'état de Peng–Robinson. L'introduction d'un seul paramètre de composant pur permet la reproduction exacte des données de pression de vapeur pour une grande variété de substances. Les composés polaires ou non polaires, associés ou non associés, sont tous bien représentés par l'équation d'état cubique PRSV. La règle de mélange classique à un paramètre binaire permet la corrélation des données d'équilibre liquide–vapeur pour une grande variété de systèmes binaires. C'est seulement pour les systèmes formés d'un composant polaire (associé ou non) et d'un hydrocarbure saturé, que les résultats sont moins bons que ceux obtenus avec les fonctions classiques de l'énergie de Gibbs d'excès.

The use of a single equation of state to reproduce the thermodynamic properties of both pure compounds and mixtures (in vapor or liquid phases) has been one of the most elusive research goals of thermodynamicists for over a century. Since van der Waals (1873) proposed his well known cubic equation of state, the number of publications in the subject has increased exponentially. With the advent of computers the use of analytical expressions to interpolate, extrapolate and even predict thermodynamic information has become of increasing importance for process design and for modelling of process operation. In principle, all required thermodynamic information of a mixture of given composition may be obtained from an equation of state valid at the temperature of interest in all the composition range and from the ideal gas state to the prevailing pressure. The impossibility of finding such a general equation of state applicable in a wide temperature range to mixtures containing nonpolar, polar and associating compounds stimulated the development of dual methods. In these methods, a model for the excess Gibbs energy of the mixture and independent information on pure compound vapor pressures are used for the liquid phase while the use of an equation of state is reserved for the vapor phase where non-idealities are less severe. For most practical purposes, the use of a single equation of state to compute phase equilibria has been limited to systems containing nonpolar or slightly polar compounds at not too low reduced temperatures of the compounds. Due to their simplicity, cubic equations of state have been popular for this kind of system. A good review of recent developments has been presented by Vidal (1983) and some limitations of the general cubic equation of state have been discussed by Vidal and Vera (1984).

New attempts to extend the applicability of cubic equations of state have been presented by Mathias (1983), Mathias and Copeman (1983), Soave (1984) and by Gibbons and Laughton (1984).

In this work we present a complete overview of the results that can be obtained with a modified Peng–Robinson equation of state, called the PRSV equation from here on. A detailed discussion of the considerations taken into account in the construction of the PRSV equation is given elsewhere (Stryjek and Vera, 1986). Although in many respects the modifications introduced in the PRSV equation follow ideas of previous workers in the field, differences in the details are significant enough to produce a definite improvement with respect to other versions of cubic equations of state. Vapor pressures of nonpolar, polar or associating compounds may be reproduced down to 1.5 kPa with accuracy comparable to the Antoine equation. Vapor–liquid equilibria of many binary systems are well represented with standard one-binary-parameter mixing rules. The cases for which the use of two binary parameters is required are identified. These cases will be treated with more detail in a following publication.

The PRSV equation of state

Peng and Robinson (1976) proposed a cubic equation of state of the form

$$P = \frac{RT}{v - b} - \frac{a}{v^2 + 2bv - b^2} \quad (1)$$

with

$$a = (0.457235 R^2 T_c^2 / P_c) \alpha \quad (2)$$

and

$$b = 0.077796 R T_c / P_c \quad (3)$$

For α , the form proposed by Soave (1972) was used

$$\alpha = [1 + \kappa (1 - T_R^{0.5})]^2 \quad (4)$$

where κ was considered to be a function of the acentric factor ω only.

In this work we retain Equations (1) to (4). After a careful examination of the deviations in the calculated vapor pres-

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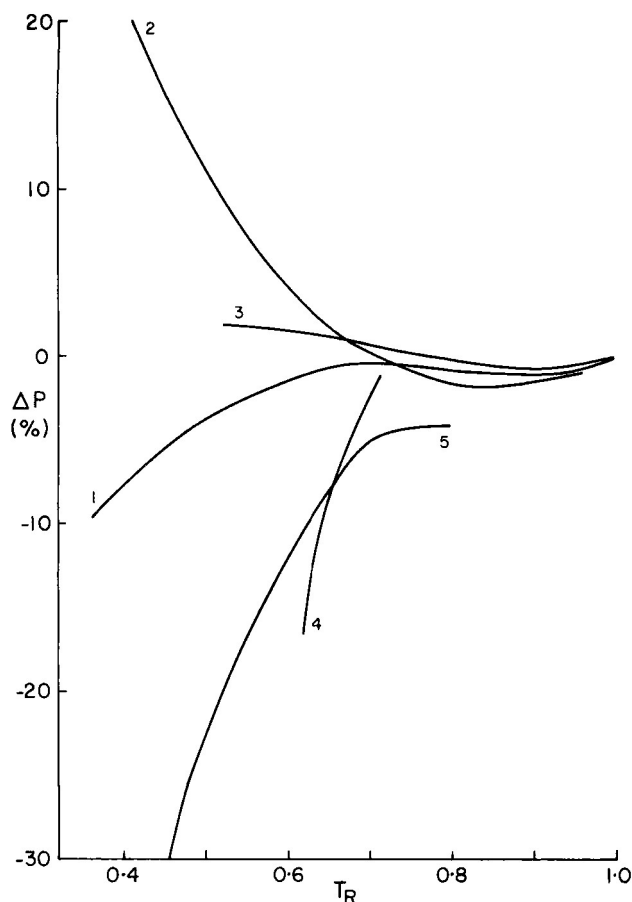


Figure 1 — Percent deviations in vapor pressures calculated with the Peng–Robinson equation as a function of reduced temperature for some typical compounds: 1. oxygen; 2. water; 3. acetone; 4. 1-butanol and 5. hexadecane.

tures at the low reduced temperatures for compounds with a wide range of acentric factors, the functional dependence of κ was modified. Figure 1 presents typical percent deviations between experimental and calculated vapor pressures as a function of reduced temperature given by the Peng–Robinson equation. It may be observed that errors are large at all temperatures for compounds with large acentric factors, even for nonpolar compounds such as hexadecane, and that the error increases rapidly at low reduced temperatures for all compounds. A major improvement is obtained with the following simple expression for κ .

$$\kappa = \kappa_0 + \kappa_1 (1 + T_R^{0.5}) (0.7 - T_R) \dots \dots \dots (5)$$

with

$$\begin{aligned} \kappa_0 = & 0.378893 + 1.4897153\omega - 0.17131848\omega^2 \\ & + 0.0196554\omega^3 \dots \dots \dots (6) \end{aligned}$$

and κ_1 , being an adjustable parameter characteristic of each pure compound. Table 1 gives the values of T_c , P_c , ω and κ_1 for over ninety compounds of industrial interest. For water and alcohols, Equation (5) with the value of κ_1 given in Table 1 applies from low reduced temperature up to the critical point. For all other compounds, slightly better results are obtained using $\kappa_1 = 0$ for reduced temperatures above 0.7.

Typical per cent vapor pressure deviations, obtained with the PRSV equation are presented in Figure 2. The change in scale between Figures 1 and 2 should be observed. As it has been discussed elsewhere (Stryjek and Vera, 1986), results

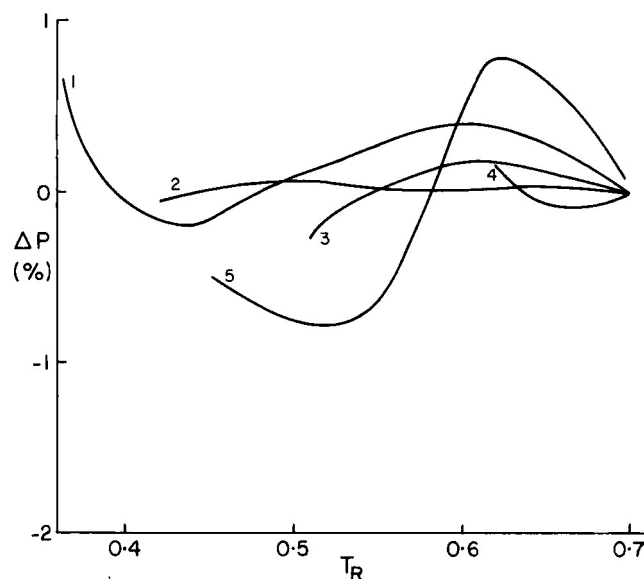


Figure 2 — Percent deviations in vapor pressures calculated with the PRSV equation as a function of reduced temperature for some typical compounds; 1. oxygen; 2. water; 3. acetone; 4. 1-butanol and 5. hexadecane.

obtained with the PRSV equation are better than those obtained by Mathias (1983), Soave (1984) and Gibbons and Laughton (1984) for the compounds included in their studies. Maximum deviations in vapor pressure calculations obtained with the PRSV equation are rarely greater than 1% and average absolute deviations are typically of the order of 0.2 to 0.3%. However, it is not recommended to use the equation at temperatures below the minimum temperature reported in Table 1 for each compound.

In this work we have reevaluated the acentric factors of some pure compounds using the best values available of saturation pressures, critical temperature and critical pressure. Values of κ_1 were then determined using equations (5) and (6) for κ in the correlation of low reduced temperature vapor pressure data. Thus, values presented in Table 1 are internally consistent and should be used together.

Due to the totally empirical nature of κ_1 , no correlation was found for it in terms of pure compound properties. For hydrocarbons and slightly polar compounds, values of κ_1 are mostly positive and smaller than 0.1. For water and ammonia, κ_1 values are small and negative. For acetic acid and methanol larger negative values of κ_1 are required. However, for higher alcohols, large positive values of κ_1 were obtained. To some extent the value of κ_1 is affected by the accuracy of the critical data. For hexadecane, for example, the recommended value of the critical pressure is 14 atm (API, 1975). Using values of the critical pressure of 14.25 atm and of 13.75 atm the corresponding values of κ_1 are 0.0095 and 0.0536 with almost the same root mean square deviations in calculated vapor pressures. As discussed below, for some compounds the values of the critical pressure and the critical temperature had to be estimated from group methods. These values are only approximate and determine the value of κ_1 . For these reasons, it is important to keep in mind the need of using the values of parameters of Table 1 without changes.

Fugacity coefficients at supercritical conditions

The method used to determine the parameters κ_0 and κ_1 at

TABLE I
Pure Compound Parameters and Per Cent Deviation in Saturation Pressures

	NP	T_c, K	P_c, kPa	ω	κ_1	T, K range	$T_R \leq 0.7$		$T_R \geq 0.7$		References
							NP	$\Delta P, \%$	NP	$\Delta P, \%$	
Inorganic											
Nitrogen	32	126.200	3400.	0.03726	0.01996	64–126	13	0.148	19	0.238	15
Oxygen	50	154.77	5090.	0.02128	0.01512	56–154	27	0.226	23	0.611	36
Carbon dioxide	23	304.21	7382.43	0.22500	0.04285	218–304	0	—	23	0.544	13
Ammonia	22	405.55	11289.52	0.25170	0.00100	195–400	10	0.105	12	0.120	36
Water	48	647.286	22089.75	0.34380	−0.06635	274–623	30	0.033	18	0.290	26
Hydrogen chloride	12	324.60	8308.57	0.12606	0.01989	159–309	6	0.852	6	1.237	31,34
Organic Hydrocarbons											
Methane	50	190.555	4595.	0.01045	−0.00159	92–190	21	0.109	29	0.458	14
Ethane	26	305.43	4879.76	0.09781	0.02669	120–293	20	0.280	6	0.472	17
Propene	47	365.57	4664.55	0.14080	0.04400	140–365	25	0.587	22	0.241	16
Propane	30	369.82	4249.53	0.15416	0.03136	128–363	23	0.782	7	0.405	17
Butane	27	425.16	3796.61	0.20096	0.03443	182–413	19	0.545	8	0.278	17
Pentane	28	469.70	3369.02	0.25143	0.03946	196–453	19	0.783	9	0.251	17
Neopentane	31	433.75	3196.27	0.19633	0.04303	259–433	9	0.089	22	0.277	23
Hexane	32	507.30	3012.36	0.30075	0.05104	232–503	17	1.106	15	0.823	17
Heptane	29	540.10	2735.75	0.35022	0.04648	254–533	14	0.885	15	0.417	17
Octane	32	568.76	2486.49	0.39822	0.04464	258–563	15	0.546	17	0.363	17
Nonane	31	594.56	2287.90	0.44517	0.04104	292–563	14	0.533	17	0.521	17
Decane	32	617.50	2103.49	0.49052	0.04510	310–563	13	0.618	18	0.900	17
Undecane	16	638.73	1965.69	0.53631	0.02919	348–499	9	0.370	7	0.157	17
Dodecane	27	658.2	1823.83	0.57508	0.05426	312–520	19	1.030	8	0.131	17
Tridecane	25	675.8	1722.51	0.62264	0.04157	336–540	17	0.671	8	0.350	17
Tetradecane	26	691.8	1621.18	0.66735	0.02686	345–559	17	0.768	9	0.804	17
Pentadecane	24	706.8	1519.86	0.70694	0.01892	337–577	14	0.513	9	0.987	17
Hexadecane	33	720.6	1418.54	0.74397	0.02665	324–594	23	0.646	10	0.844	17
Heptadecane	23	733.4	1317.21	0.76976	0.04048	401–610	13	0.840	10	1.079	17
Octadecane	23	745.2	1215.89	0.79278	0.08291	413–625	13	1.417	10	0.790	17
Cyclohexane	24	553.64	4075.	0.20877	0.07023	280–553	13	0.363	11	0.231	25
Bicyclohexyl	23	731.4	2563.50	0.39361	0.01805	424–577	14	0.597	9	1.261	31,37
Benzene	32	562.16	4898.	0.20929	0.07019	279–543	17	0.541	15	0.319	17
Toluene	34	591.80	4106.	0.26323	0.03849	286–583	17	0.363	17	0.346	17
Ethylbenzene	34	617.20	3606.	0.30270	0.03994	306–603	16	0.303	18	0.400	17
<i>p</i> -Xylene	34	616.23	3511.	0.32141	0.01277	308–603	16	0.317	18	0.584	17
Indane	25	684.90	3950.	0.31000	0.01173	355–482	24	0.416	1	0.027	9
<i>n</i> -Propylbenzene	34	638.32	3200.	0.34513	0.02715	324–633	15	0.342	19	0.113	17
1,2,3-Trimethylbenzene	34	637.25	3127.	0.39970	−0.01384	330–633	15	0.184	19	0.206	17
Naphthalene	16	748.35	4050.93	0.30295	0.03297	360–523	16	0.432	0	—	17
1-Methyl-naphthalene	24	766. **	3566.60	0.37666	−0.01842	424–593	16	0.133	8	0.359	38
2-Methyl-naphthalene	29	761.	3505.81	0.37119	−0.01639	424–639	16	0.260	13	1.115	38
Biphenyl	8	769.15	3120.78	0.38095	0.11487	293–366	8	0.444	0	—	21,31
Diphenylmethane	30	770.2	2857.34	0.43724	0.05955	425–647	18	0.751	12	1.194	31,37
9,10-Dihydro-phenanthrene	17	774.7 **	1314.17	0.54678	−0.01393	437–553	15	1.099	2	1.770	31,38
Ketones											
Acetone	45	508.1	4696.	0.30667	−0.00888	259–553	30	0.125	15	0.435	5,7
Butanone	43	536.78	4207.	0.32191	0.00554	316–553	18	0.096	25	0.796	7
2-Pentanone	17	561.08	3694.	0.34719	0.01681	336–385	17	0.076	0	—	7
3-Pentanone	18	561.46	3729.	0.34377	0.03558	330–384	18	0.080	0	—	7
Methylbutanone	19	555.	3790.	0.31314	0.04113	329–377	19	0.058	0	—	7
2-Hexanone	30	587.	3320.	0.39385	0.00984	308–428	26	0.328	4	0.101	7
3-Hexanone	18	582.82	3319.	0.37931	0.02321	349–407	18	0.081	0	—	7
Dimethylbutanone	30	567.	3470.	0.32293	0.04005	289–405	28	0.363	2	0.038	7
2-Heptanone	30	611.5	2990.	0.42536	0.02731	328–452	25	0.319	5	0.030	7
5-Nonanone	28	640.	2329.	0.51374	0.02002	358–485	19	0.235	9	0.129	7

TABLE 1 (cont'd.)

	<i>NP</i>	<i>T_c</i> , K	<i>P_c</i> , kPa	ω	κ_1	<i>T</i> , K range	<i>T_R</i> ≤ 0.7		<i>T_R</i> ≥ 0.7		References
	<i>NP</i>						<i>NP</i>	ΔP , %	<i>NP</i>	ΔP , %	
Alcohols											
Methanol	43	512.58	8095.79	0.56533	−0.16816	288–485	21	0.274	22	0.915	1,4,6
Ethanol	47	513.92	6148.	0.64439	−0.03374	293–485	23	0.463	24	0.949	1,6
1-Propanol	19	536.71	5169.55	0.62013	0.21419	333–378	17	0.196	2	0.076	1
2-Propanol	17	508.40	4764.25	0.66372	0.23264	325–362	12	0.099	5	0.082	1
1-Butanol	18	562.98	4412.66	0.59022	0.33431	352–399	14	0.054	4	0.067	1
2-Butanol	17	535.95	4248.52	0.58254	0.39045	341–380	13	0.016	4	0.017	1
2-Methyl-1-Propanol	18	547.73	4295.12	0.59005	0.37200	343–389	14	0.052	4	0.049	1
2-Methyl-2-Propanol	16	506.15	3971.90	0.61365	0.43099	330–363	10	0.108	6	0.006	1
1-Pentanol	35	588.15	3909. *	0.57839	0.36781	348–514	15	0.073	20	1.124	1
1-Hexanol	29	591.23**	3468.15*	0.77526	−0.00237	313–438	23	0.280	6	0.809	39
1-Octanol	46	684.8 **	2860.00*	0.32420	0.82940	328–554	30	0.180	16	3.302	4,6
1-Decanol	32	717.84**	2394.87*	0.38355	0.80898	313–503	30	0.705	2	0.079	39
Ethers											
Dimethyl Ether	34	400.1	5240.	0.18909	0.05717	183–503	14	0.452	20	0.720	8,22,27
Methyl Ethyl Ether	12	437.8	4410.	0.23479	0.16948	273–428	5	0.303	7	1.862	8,18
Methyl <i>n</i> -Propyl Ether	22	476.25	3801.	0.27215	0.02300	254–333	22	0.330	0	—	8
Methyl <i>i</i> -Propyl Ether	20	464.48	3762.	0.26600	0.04123	250–325	20	0.154	0	—	8
Methyl <i>n</i> -Butyl Ether	29	512.78	3371.	0.31672	0.01622	266–367	27	0.270	2	0.027	8
Methyl <i>t</i> -Butyl Ether	16	497.1	3430.	0.26746	0.05129	288–351	15	0.068	1	0.112	8
Ethyl <i>n</i> -Propyl Ether	29	500.23	3370.	0.33612	−0.01668	261–359	26	0.495	3	0.178	8
Di- <i>n</i> -Propyl Ether	25	530.6	3028.	0.37070	−0.03162	293–388	21	0.351	4	0.391	8
Di- <i>i</i> -Propyl Ether	20	500.32	2832.	0.33168	0.03751	285–365	16	0.160	4	0.063	8
Methyl Phenyl Ether	17	645.6	4250.	0.34817	0.01610	383–427	17	0.122	0	—	8
Various											
Nitromethane	10	588.	6312.49	0.34700	−0.10299	244–374	10	1.189	0	—	31,33
Acetonitrile	35	545.5	4830.	0.33710	−0.13991	280–530	20	0.260	15	5.632	19,28,30,35
Acetic Acid	27	592.71	5786.	0.45940	−0.19724	304–415	26	0.379	1	0.023	10
Dimethylformamide	7	660.07*	5240.66*	0.26600	0.18999	303–363	7	0.748	0	—	20
2-Methoxyethanol	19	574.39*	5348.38*	0.65629	−0.42503	329–397	19	0.275	0	—	20
1-Propylamine	12	497.	4742.	0.28037	0.14326	296–351	11	0.112	1	0.222	11,20
2-Propylamine	13	476.	5066.20	0.28530	0.06001	277–334	12	0.334	1	0.031	11,20
2-Methoxypropionitrile	14	636.11*	3602.55*	0.47656	−0.09508	293–436	14	0.733	0	—	32
2-Methyl-2-Propylamine	12	483.9	3840.	0.27417	0.13440	292–348	10	0.135	2	0.429	11,20
Tetrahydrofuran	15	540.1	5190	0.22550	0.03961	296–373	15	0.136	0	—	11,20
Pyridine	15	620.	5595.26*	0.23716	0.06946	340–426	15	0.116	0	—	2,20
Furfural	34	652.48*	4345.45*	0.39983	−0.03471	329–434	34	3.067	0	—	20
<i>N</i> -Methylpyrrolidone	15	719.33*	4057.72*	0.34478	0.11367	373–478	15	1.403	0	—	40
Hexafluorobenzene	23	516.7	3273.	0.39610	0.02752	278–387	18	0.282	5	0.021	2,20
Nitrotoluene	15	743.	3207.	0.42200	−0.00901	417–499	15	0.133	0	—	12
<i>m</i> -Cresol	25	705.15	4559.58	0.44492	0.24705	401–594	20	1.439	5	2.099	20,29,31
Thianaphthene	30	752.	3880.71	0.29356	0.06043	424–631	15	0.367	15	1.924	37

*Estimated by group contribution method.

**Obtained from optimum fit (see Table 3).

References to Table 1. 1) Ambrose and Sprake (1970), 2) Ambrose et al. (1970), 3) Ambrose et al. (1974a), 4) Ambrose et al. (1974b), 5) Ambrose et al. (1974b), 6) Ambrose et al. (1975a), 7) Ambrose et al. (1975b), 8) Ambrose et al. (1976), 9) Ambrose and Sprake (1976), 10) Ambrose et al. (1977), 11) Ambrose (1978), 12) Ambrose and Gundry (1980), 13) Angus et al. (1976), 14) Angus et al. (1978), 15) Angus et al. (1979), 16) Angus et al. (1980), 17) API 44 Tables, 18) Berthoud and Brum (1924), 19) Brown and Smith (1954), 20) Boublik et al. (1973), 21) Chipman and Peltier (1929), 22) Cardoso and Bruno (1923), 23) Dawson et al. (1973), 24) Gmehling and Onken (1980), 25) Huggill and McGlashan (1978), 26) Keenan et al. (1978), 27) Kennedy et al. (1941), 28) Mousa (1981), 29) Nasir et al. (1980), 30) Putnam et al. (1965), 31) Reid et al. (1977), 32) Stryjek et al. (1978), 33) Stulle (1947a), 34) Stulle (1947b), 35) Trejo and McLure (1979), 36) Vargaftik (1975), 37) Wiecek and Kobayashi (1980), 38) Wiecek and Kobayashi (1981), 39) Wilhoit and Zwolinski (1973), 40) Yarym-Agaev et al. (1980).

subcritical temperatures was the standard one, i.e., to obtain equality of fugacities of the saturated phases at a given temperature. For the supercritical region, values of the parameters, or their temperature dependence, may be obtained from the fitting of volumetric data.

Since the Peng-Robinson equation of state gives satisfactory results for compounds of industrial interest (nitrogen, carbon dioxide, methane, etc.) in the supercritical region, the temperature dependence of κ seems not to be required in this region. In fact, preliminary studies showed that there is no advantage in using the PRSV equation

with $\kappa_1 \neq 0$ in this region. Thus, for the supercritical region, $T_R \geq 1$, the use of $\kappa = \kappa_0$ is recommended for all compounds.

For all practical purposes both the Peng-Robinson equation and PRSV equation represent fugacities at the supercritical state with the same precision. The greatest errors are produced in the critical region of the compound. On the average, the differences in fugacity coefficients computed with both equations are smaller than 0.01%, the PRSV being slightly better in most cases. Table 2 compares the performance of both equations for the prediction of fugac-

TABLE 2
Per Cent Deviations in Supercritical Fugacities of Methane Calculated with the
Peng–Robinson Equation (PR) and the PRSV Equation

P, MPa T, K (T _R)		0.5	1.0	5.0	10.	20.	30.
195. (1.0233)	PR	0.591	1.141	4.136	3.782	5.310	7.522
	PRSV	0.590	1.140	4.128	3.765	5.291	7.502
240. (1.2595)	PR	0.424	0.824	3.319	4.854	5.246	6.469
	PRSV	0.421	0.819	3.288	4.784	5.126	6.332
290. (1.5219)	PR	0.288	0.561	1.304	3.674	4.830	5.704
	PRSV	0.284	0.554	1.287	3.602	4.698	5.535
380. (1.9942)	PR	0.158	0.311	1.335	2.258	3.483	4.428
	PRSV	0.155	0.304	1.304	2.196	3.364	4.275
480. (2.5190)	PR	0.094	0.186	0.837	1.492	2.517	3.405
	PRSV	0.092	0.181	0.811	1.443	2.426	3.280

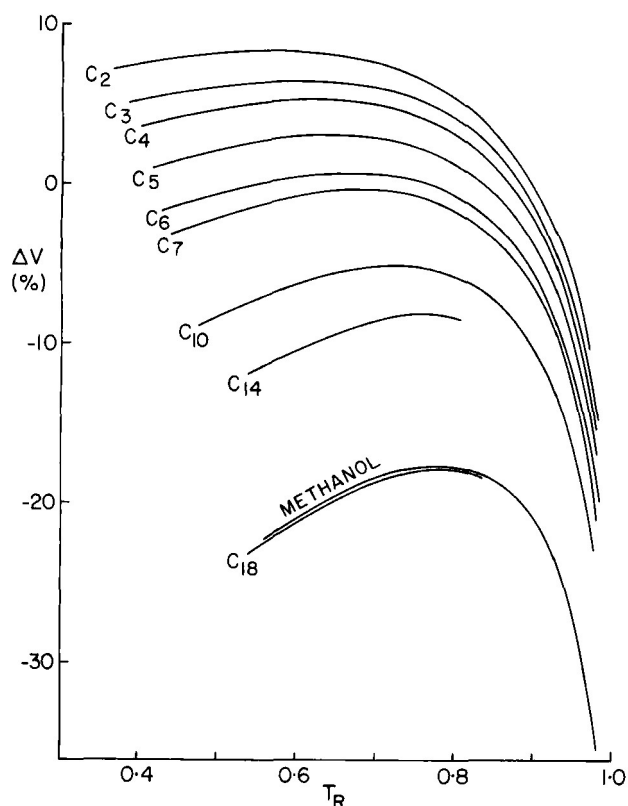


Figure 3 — Percent deviations in liquid molar volumes at saturation calculated with the PRSV equation. (C_n represents the alkane $C_n H_{2n+2}$).

ities of methane in the supercritical region. The fugacities for methane were taken from Angus et al. (1978).

Liquid molar volumes and second virial coefficients from the PRSV equation of state

As discussed by Abbott (1979), a cubic equation of state should not be expected to reproduce all thermodynamic properties accurately. In this work we have given preference to the representation of vapor pressures of pure compounds with the aim of reproducing vapor–liquid equilibria in mixtures. As shown by Martin (1979) the prediction of liquid molar volumes by cubic equations of state may be improved

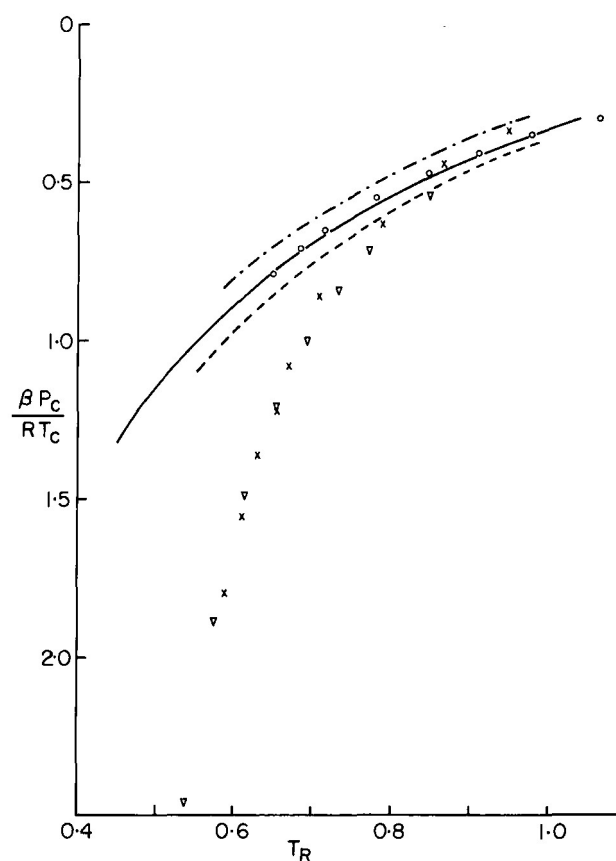


Figure 4 — Reduced second virial coefficient as a function of reduced temperature. Values recommended by Dymond and Smith (1980) are represented by points, lines represent values calculated with the PRSV equation. Compounds: O, — ethane; x, — acetone; ∇, --- water.

by the introduction of an additional pure component parameter to produce a translation of the isotherms. Peneloux et al. (1982) have shown that using the proper mixing rules, the introduction of the new parameter does not affect the prediction of vapour pressures or of equilibria. Although in this work we have not attempted to introduce a translation parameter to improve the prediction of liquid molar volumes with the PRSV equation, the possibility of doing so is kept open. For the purposes of this work it suffices to note that the predicted liquid molar volumes are in reasonable agree-

TABLE 3
Effect of the Estimated Critical Temperature on the Representation of Vapor Pressure Data. a) T_c Estimated by Ambrose Method. b) T_c for Optimum Fit

Compound (P_c , kPa)	T_c , K	ω	κ_1	$\overline{\Delta P}$ (%)
1-methylnaphthalene (3564.6)	a)772. b)766.	0.3400 0.3767	0.0611 -0.0184	0.247 0.133
9,10-dihydrophenanthrene (1314.2)	a)774. b)774.7	0.5434 0.5468	0.0344 -0.0139	1.716 1.099
1-hexanol (3468.1)	a)610. b)591.2	0.5796 0.7753	0.3761 -0.0024	1.865 0.386
1-octanol (2860.0)	a)652. b)684.8	0.5936 0.3242	0.3881 0.8294	1.627 0.180
1-decanol (2394.9)	a)700. b)717.8	0.5277 0.3836	0.5807 0.8084	1.908 0.666

TABLE 4
Effect of the Estimated Critical Pressure on the Representation of Vapor Pressure Data. a) P_c Estimated by Ambrose Method. b) P_c for Optimum Fit

Compound (T_c , K)	P_c , kPa	ω	κ_1	$\overline{\Delta P}$ (%)
1-methylnaphthalene (772.0)	a)3564.6 b)3792.1	0.3400 0.3666	0.0611 -0.0215	0.247 0.122
9,10-dihydrophenanthrene (774.0)	a)1314.2 b)1987.4	0.5434 0.7231	0.0344 -0.5130	1.716 0.874
1-hexanol (610.0)	a)3468.1 b)4616.3	0.5796 0.7038	0.3761 0.0183	1.865 0.316
1-octanol (652.0)	a)2860.0 b)2238.9	0.5936 0.4872	0.3881 0.6965	1.627 0.141
1-decanol (700.0)	a)2394.9 b)2045.9	0.5277 0.4593	0.5807 0.7630	1.908 0.564

ment with actual data for most compounds over a wide temperature range. In addition, the very systematic character of the deviations suggested that a correction of the type described above is perfectly possible. Typical deviations of calculated liquid volumes with respect to those obtained with the Rackett equation in the version of Spencer and Adler (1978) are shown in Figure 3. Excluding the critical region, where deviations are large, low molecular weight paraffins show deviations of less than 8%. From butane to heptane, deviations are within $\pm 6\%$ up to $T_R = 0.85$. For higher hydrocarbons, deviations become larger and negative. A similar trend was observed for cyclic hydrocarbons. Polar compounds also showed a regular behavior of the deviations as a function of reduced temperature. Relatively great average deviations were found for acetone (-12%), methanol (-20%) and acetic acid (-30%). Higher alcohols, on the other hand, present only small deviations, typically of the order of -2 to -4% .

Cubic equations of state, when expanded in terms of volume, give the following expression for the reduced second virial coefficient.

$$\frac{\beta P_c}{RT_c} = \Omega_b - \Omega_a \frac{\alpha}{T_R} \dots \dots \dots (7)$$

Figure 4 presents a comparison of calculated and smoothed experimental values for ethane, acetone and water. Smoothed experimental values were taken from the com-

pilation of Dymond and Smith (1980). The representation of the second virial coefficient for ethane is typical for non-polar or slightly polar compounds. Larger errors are found for polar and associating compounds. The performance of the PRSV equation in this respect, is quite typical of cubic equations of state. It may be improved, if desired, by a method similar to that proposed by Martin (1984).

The use of the PRSV equation when P_c and T_c are unknown

While saturation pressure data at temperatures close to the normal boiling point are usually available for compounds of industrial interest, values of the critical temperature and pressure may be unknown. This is one of the main limitations encountered for a wide applicability of generalized equations of state.

Assuming that the normal boiling point of the compound in question is known, the values of the critical temperature and of the critical pressure can be estimated by group contribution methods. We have studied the performance of the PRSV equation with values of T_c and P_c evaluated by the Lydersen (1955), Ambrose (1980) and Klinciewicz and Reid (1984) methods. In general, use of the Ambrose method resulted in values of ω and κ_1 that allowed a better representation of the vapor pressure data. Once the values of T_c

TABLE 5
Average Error in Calculated Pressure, kPa, and Vapor-phase Mole Fraction

System	NI	NP	Temperature range, K.	PRSV		Wilson*		NRTL*		UNIQUAC*		Reference
				$\overline{\Delta P}$	$\overline{\Delta y} \cdot 10^2$	$\overline{\Delta P}$	$\overline{\Delta y} \cdot 10^2$	$\overline{\Delta P}$	$\overline{\Delta y} \cdot 10^2$	$\overline{\Delta P}$	$\overline{\Delta y} \cdot 10^2$	
1. Benzene/cyclohexane	12	113	283.15–392.45	0.23	0.31	0.16	0.17	0.17	0.17	0.18	0.17	1a
2. Benzene/hexane	9	109	298.15–343.15	0.13	0.80	0.14	0.59	0.14	0.60	0.14	0.60	1b
3. Cyclohexane/hexane	1	7	343.15	0.24	0.41	0.44	0.36	0.39	0.37	0.39	0.37	1c
4. Benzene/hexadecane	5	69	298.15–353.15	0.19	—	0.08	—	0.09	—	0.08	—	1d
5. Benzene/biphenyl	3	46	318.15–338.15	0.08	0.03	1.87	0.02	0.60	0.01	2.49	0.03	1e
6. Hexane/hexadecane	5	50	293.15–333.15	0.10	—	0.02	—	0.02	—	0.02	—	1f
7. Acetone/benzene	7	78	298.15–323.15	0.32	0.88	0.21	0.50	0.21	0.51	0.21	0.50	1g
8. Acetone/cyclohexane	5	70	298.15–328.15	1.34	2.16	0.55	0.53	0.53	0.69	0.58	0.72	1h
9. Acetone/hexane	7	88	253.15–318.15	1.06	2.46	0.47	0.81	0.43	0.83	0.41	0.87	1i
10. Benzene/1-butanol	2	16	298.15–318.15	0.58	0.80	0.42	1.04	0.48	1.38	0.47	1.37	1j
11. Cyclohexane/1-butanol	8	100	298.15–383.15	3.75	1.95	1.80	1.52	1.82	1.52	1.91	1.57	1k
12. Hexane/1-butanol	1	9	298.15	1.78	0.99	2.23	0.59	2.24	0.58	2.25	0.58	1l
13. Water/methanol	11	137	298.15–373.15	0.57	1.53	0.51	1.01	0.42	1.00	0.45	1.06	1m
14. Ammonia/water**	9	82	222.04–394.26	4.50	0.47	—	—	—	—	—	—	2

*Calculated results obtained from the same reference of the data

** $\overline{\Delta P}$ expressed in per cent deviation

References for Table 5. 1) Gmehling et al (1980), a) Vol. 1, 6a, pp. 204, 206, 207, 211, 214, 217, 220, 221, 223, 229, 237, 239; b) Vol. 1, 6a, pp. 534, 542–546, 548, 556, 558; c) Vol. 1, 2a, p. 276; d) Vol. 1, 6b, pp. 448–452; e) Vol. 1, 7, pp. 324–326; f) Vol. 1, 2a, pp. 614–618; g) Vol. 1, 3–4, pp. 194, 199–203, 208; h) Vol. 1, 3–4, pp. 210–212, 214, 216; i) Vol. 1, 3–4, pp. 222–224, 227–230; j) Vol. 1, 2b, pp. 176, 177; k) Vol. 1, 2b, pp. 184–187, 189–192; l) Vol. 1, 2b, p. 201; m) Vol. 1, 1, pp. 38, 39, 41, 42, 49, 55–57, 72, 73, and la, p. 49. 2) Macriss, R. A. et al. (1964).

and P_c were obtained from the Ambrose method, ω was calculated from smoothed vapor pressure data at $T_R = 0.7$ or when such data was not available, from Edmister's equation (Reid et al., 1977).

$$\omega = \frac{3}{7} \left(\frac{T_b}{T_c - T_b} \right) \log_{10} P_c \text{ (atm)} - 1.000 \quad (8)$$

and κ_0 was then calculated from Equation (6). The value of κ_1 was then obtained by fitting the available vapor pressure data. The advantage of this procedure is that assures internal consistency of all parameters and that κ_1 absorbs to some extent the effect of the uncertainties in T_c or P_c .

It was found, however, that for some few compounds for which group methods represent a large extrapolation, even the optimum value of κ_1 produced deviations in vapor pressures larger than those obtained for compounds for which critical constants had been measured experimentally. In addition, it was observed that changes in the critical parameters, well within the uncertainty of the extrapolation by group methods, produced, in some cases, significant improvement in the quality of the fit of the available vapor pressure data in the low reduced temperature region. Table 3 shows the effect of a variation of the critical temperature on the quality of the fit for some selected compounds for which estimated critical constants were required. Table 4 shows the effect of a variation of the critical pressure for the same compounds. Values reported in Table 1 for the critical parameters are those that were found to represent a good compromise between the uncertainty of the estimate by group methods and the quality of the reproduction of low reduced temperature vapor pressure data. All cases for which this was the case are specifically indicated in Table 1. For some compounds even this method produced errors larger than the norm. In the case of furfural, errors may be attributed to thermal decomposition at temperatures around its normal boiling point. In the case of acetonitrile there may be a systematic error in the data at high reduced temperatures.

Vapor-liquid equilibria calculations for binary mixtures with the PRSV equation of state

In this work we study the performance of the PRSV equation using only one binary parameter for the calculation of binary vapor-liquid equilibria. Results are compared with those obtained using a dual approach, i.e. an expression for the excess Gibbs energy of the liquid phase and an equation of state for the gas phase.

In order to obtain a better perspective of the relative performance of the PRSV equation, it is necessary to consider the number of adjustable parameters involved in both methods. Values of T_c , P_c and ω , for each pure compound, are used by both methods. Dual approaches usually include these parameters to obtain vapor phase fugacities from the truncated virial equation of state. In addition, the PRSV equation, as used here, will include two pure compound parameters (κ_1) and one binary parameter. Dual approaches require at least six pure compound parameters (three Antoine constants for each pure compound, for example) and typically two binary parameters for the expression of the excess Gibbs energy. Thus, calculations with the PRSV equation involve three adjustable parameters while dual methods use at least eight adjustable parameters.

For the PRSV equation we have initially used the conventional mixing rules

$$b = \sum x_i b_i \quad (9)$$

and

$$a = \sum x_i x_j a_{ij} \quad (10)$$

with

$$a_{ij} = (a_i a_j)^{0.5} (1 - k_{ij}) \quad (11)$$

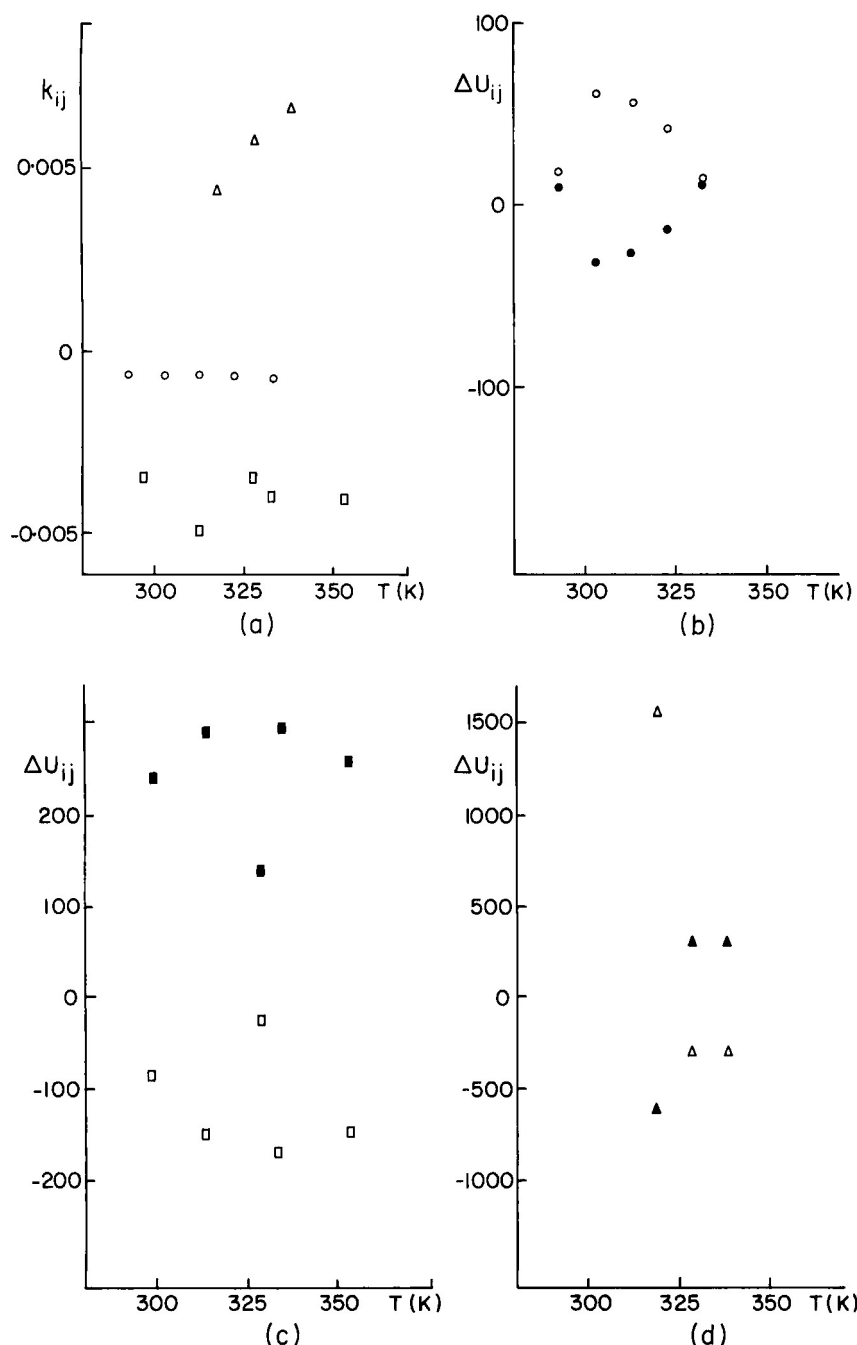


Figure 5 — Binary parameters for the systems: ○, hexane/hexadecane; □, benzene/biphenyl. (a) k_{12} for the PRSV equation; (b), (c) and (d), ΔU_{12} (open symbol) and ΔU_{21} (full symbol) for the UNIQUAC equation (Gmehling et al., 1980).

The expression for the fugacity coefficient of a component i in a mixture is then the same as for the Peng–Robinson equation, namely

$$\ln \psi_i = \frac{b_i}{b} (z - 1) - \ln(z - B) - \frac{A}{2\sqrt{2}B} \times \left(\frac{2 \sum_k x_k a_{ik}}{a} - \frac{b_i}{b} \right) \ln \frac{z + (1 + \sqrt{2})B}{z + (1 - \sqrt{2})B} \dots (12)$$

Results obtained with the one-binary-parameter PRSV equation, designated as PRSV-1, are compared in Table 5 with those obtained with dual methods. For convenience we have selected representative systems for which results with dual methods have been previously calculated by Gmehling et al. (1980). These include the commonly used expressions for the excess Gibbs energy of Wilson (1964), the NRTL three-parameter equation of Renon and Prausnitz (1968) and the UNIQUAC equation of Abrams and Prausnitz (1975).

The systems presented in Table 5 represent various classes of mixtures of nonelectrolytes for which isothermal data are available over a wide temperature range.

The first three systems of Table 5 correspond to mixtures of nonpolar/nonpolar compounds where the molecules are of similar size but different shape. These mixtures are well represented by the one-binary-parameter PRSV-1 equation of state. Average errors are similar to those obtained with dual approaches. Values of k_{12} for each system are only slightly temperature dependent over the whole temperature range. In addition, k_{12} values present a linear temperature dependence as shown in Figure 5a. Figures 5b to 5d show the variation with temperature of UNIQUAC parameters for the same systems. While for the PRSV-1 equation one single figure suffices to present the parameters, for the UNIQUAC parameters different scales are required. Systems 4 to 6 of Table 5 represent the case of mixtures of nonpolar/nonpolar compounds with great difference in size. The difference in size is associated with large differences in

critical parameters and thus in the reduced conditions of pressure and temperature for both compounds. Again here, the PRSV-1 equation performs well and k_{12} parameters are slightly temperature dependent. Only for system 6 the temperature dependence of k_{12} is greater than for other nonpolar systems; however, it was found to be linear.

Systems 7 to 9 of Table 5 represent mixtures of polar (non-associated)/nonpolar compounds. For these systems the one-binary-parameter version of the PRSV equation performs well with aromatic nonpolar compounds but produces larger deviations than dual methods when the nonpolar compound is a saturated hydrocarbon. Detailed analysis of the deviations in calculated pressures obtained with the PRSV-1 equation showed that these are not random in nature but present a systematic behavior that could be eliminated if two binary parameters were used.

Systems 10 to 12 of Table 5 present the case of mixtures of polar (associated)/nonpolar compounds. The performance of the PRSV-1 equation is similar in all respects to the case of systems 7 to 9. Thus, it appears natural to conclude that the systematic deviations encountered with the PRSV-1 equation are not due to the associated or non-associated nature of the polar compound but the aliphatic or aromatic nature of the nonpolar compound.

System 13 exemplifies the performance of the PRSV-1 equation for polar (associated)/polar (associated) compounds. In this case deviations in pressure and vapor phase mole fraction produced by the PRSV-1 equation are only slightly greater than those obtained by dual methods. For this system, however, the deviations obtained with the PRSV-1 equation are rather random than systematic and no great improvement may be expected from the introduction of a second binary parameter. k_{12} values obtained for this system have a slight and regular temperature dependence.

Finally Table 5, shows the results obtained with the PRSV-1 equation for the system water/ammonia. Although reported deviations may seem large, the values calculated with the PRSV-1 equation are in better agreement with experimental values than those reported by Skogestad (1983).

Conclusions

A simple modification of the Peng–Robinson equation of state has been developed. The introduction of one adjustable parameter per pure compound for reduced temperatures below 0.7 has allowed the extension of the use of a cubic equation of state to the low reduced temperature region. The values of the adjustable parameter have been determined for over ninety compounds of industrial interest. These include hydrocarbons of complex molecules important in carbochemistry and in the separation of petroleum heavy ends such as bicyclohexyl, biphenyl, 9,10 dihydrophenanthrene, diphenylmethane, etc. Polar compounds such as ketones, ethers, polar aromatics; polar associating compounds such as alcohols and water; multifunctional polar compounds used as solvents in extractive distillation processes and some inorganic compounds such as ammonia, carbon dioxide, etc., have also been included in this study.

The modified form of the Peng–Robinson equation, the PRSV equation, reproduces pure compound vapor pressures with accuracy better than 1% down to 1.5 kPa.

Vapor–liquid equilibria of binary mixtures of nonpolar/nonpolar type may be represented with accuracy similar to that of standard dual methods using a single binary param-

eter. The binary parameter is slightly temperature dependent and in general follows a linear behavior with respect to temperature.

Mixtures containing polar compounds and aromatic compounds may also be represented with the use of one single binary parameter. For mixtures containing polar compounds and saturated hydrocarbons, two binary parameters are required. The use of two-parameter mixing rules will be discussed in a following publication.

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Notation

a, b	= equation of state parameters
$\frac{a_i}{a_i}$	= $(\delta n a / \delta n_i)$
A, B	= dimensionless terms, $A = Pa/(RT)^2$; $B = Pb/RT$
k	= binary parameter
n	= number of moles
NI	= number of isotherms
NP	= number of points
P	= pressure
R	= gas constant
v	= molar volume
x	= mole fraction
y	= vapor phase mole fraction
z	= compressibility factor

Greek letters

α	= function of reduced temperature and acentric factor
β	= second virial coefficient
$\frac{\Delta P}{\Delta U}$	= $\sum P_{\text{calculated}} - P_{\text{experimental}} /NP$
ΔU	= parameters for the UNIQUAC equation
$\frac{\Delta y}{\Delta y}$	= $\sum y_{\text{calculated}} - y_{\text{experimental}} /NP$
κ	= function of reduced temperature and acentric factor
κ_0	= function of acentric factor
κ_1	= pure compound parameter
ψ	= fugacity coefficient
ω	= acentric factor
Ω_a	= numerical constant. For the PRSV equation $\Omega_a = 0.457235$
Ω_b	= numerical constant. For the PRSV equation $\Omega_b = 0.077796$

Subindices

c	= at critical conditions
i, j	= compounds
R	= reduced property

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